Diagnosing Swapped Input–Output and Function-Notation Errors
Answer Key

Grade 8 / Pre-Algebra • Error Analysis

Quick Answers

1. 7	3. 1	5. 10	7. 3
2. 6	4. 15	6. —	8. —

Curriculum

Level: Grade 8 / Pre-Algebra • Error Analysis
8.F.A.1–8.F.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got 2: treated 3 as the OUTPUT and solved 4x - 5 = 3 instead of substituting x = 3*
2. *If they got 71: substituted the OUTPUT 19 for the input and computed f(19)*
3. *If they got 4: read the table backwards: hunted for 2 in the OUTPUT row and reported the month above it*
4. *If they got 10: split the input: computed f(2) + f(3) as if f distributed over addition*
4. *If they got 21: multiplied the two outputs instead, computing f(2) times f(3)*
5. *If they got 73: fed the output (28 dollars) in as the input and computed C(28)*
7. *If they got -42: evaluated h(12) instead of solving h(x) = 12*

1. $f(3)$ names the *output* produced by the input 3, so substitute 3 for x: $f(3) = 4(3) - 5 = 12 - 5 = 7$. $\boxed{f(3) = 7}$

Watch for: an answer of 2. That comes from setting $4x - 5 = 3$, which treats the given 3 as an output. It is the right work for a different question — the one asked in problem 8.

2. Here 19 is the *output*, so the equation to solve is $4x - 5 = 19$. Add 5 to both sides: $4x = 24$. Divide by 4: $x = 6$. Check by evaluating forwards: $f(6) = 4(6) - 5 = 19$. $\boxed{x = 6}$

Watch for: 71, which is $f(19) = 4(19) - 5$ — the output substituted into the input slot.

3. Priya's method searches the $g(m)$ row for the number 2 and reports the month sitting above it. That answers the question "for which month were there 2 rainy days?" — and its answer really is month 4. It is not what $g(2)$ asks. $g(2)$ means: go to the column headed $m = 2$ and read the value underneath it, which is $\boxed{g(2) = 1}$
.

Accept any wording that says Priya searched the output row instead of the input row, or that she answered "which month gives 2?" instead of "what does month 2 give?"

4. Compute the correct side first. Inside the parentheses is a single input: $2 + 3 = 5$, so $f(2 + 3) = f(5) = 4(5) - 5 = 20 - 5 = 15$. $\boxed{f(2+3) = 15}$

Jo's method gives $f(2) + f(3) = 3 + 7 = 10$, which is not 15. The notation $f(\ )$ is a *name for a rule*, not a multiplication, so it cannot be distributed across a sum. Splitting the input like this only ever works for very special functions, and $f(x) = 4x - 5$ is not one of them.

Also watch for: 21, from multiplying the two outputs, $f(2) \times f(3) = 3 \times 7$. Same misreading of $f(\ )$ as an operation on the pieces rather than a rule applied to one input.

5. $C(28)$ substitutes 28 into the *input* slot, so it answers "what does a 28-mile ride cost?" ($C(28) = 3 + 2.5(28) = 73$ dollars). Kim's \$28.00 is money, which lives in the *output* slot of C. The correct equation is

$$3 + 2.5m = 28 \quad\Longrightarrow\quad 2.5m = 25 \quad\Longrightarrow\quad m = 10.$$

Kim can ride $\boxed{m = 10}$
miles. Check: $C(10) = 3 + 2.5(10) = 28$ dollars.

6. Theo compared the two functions by scanning for the biggest number anywhere in a table, which ignores the input entirely. A comparison of functions has to happen at the *same input*. At the input 4: from the table $g(4) = 2$, and from the rule $B(4) = 2(4) - 3 = 5$. Since 2 is less than 5,

$$\boxed{g(4) < B(4)}$$

Note that the answer can flip at a different input, which is exactly why naming the input matters.

7. The first wrong line is the very first one: Marcus wrote $h(12)$. The 12 he was given is an *output*, but $h(12)$ puts it in the input slot — the swap happens before any arithmetic, so everything after it is beside the point. (His arithmetic is actually correct: $h(12)$ really is -42.)

The correct set-up puts 12 on the output side:

$$30 - 6x = 12 \quad\Longrightarrow\quad -6x = -18 \quad\Longrightarrow\quad x = 3.$$

The input is $\boxed{x = 3}$
. Check: $h(3) = 30 - 6(3) = 30 - 18 = 12$.

8. Manual review — expected reasoning. A complete answer states a usable rule and then demonstrates it. A model answer:

Rule: look at where the given number is written. If it is inside the parentheses, it is an input, so substitute and compute (evaluate). If it is on the other side of an equals sign from $f(x)$, it is an output, so write an equation and solve for x.

Applying it: "Find $f(3)$" has the 3 inside the parentheses, so substitute: $f(3) = 4(3) - 5 = 7$. "Find x when $f(x) = 3$" has the 3 opposite $f(x)$, so solve: $4x - 5 = 3$, giving $4x = 8$ and $x = 2$.

Why the answers differ: the two questions travel through the rule in opposite directions — one goes input → output, the other output → input — so there is no reason for them to give the same number. A student who gets 7 for both, or 2 for both, has collapsed the two directions into one.

Accept any rule a classmate could actually follow (position of the number, "inside the parentheses vs. across the equals sign", input → output vs. output → input) that produces 7 and 2 respectively. Do not accept "look at the wording" with no test to apply.